

MMRepAgent: Explainable stock earnings forecasting via multimodal report agent framework

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ABSTRACT

Stock return forecasting plays a critical role in quantitative investment but remains challenging for ordinary investors due to the difficulty of analyzing complex financial data and unstructured news content. While large language models (LLMs) offer interactive capabilities, they often require users to possess substantial financial expertise to issue effective prompts. To address these limitations, we propose **MMRepAgent**, an automated system that enables retail investors to generate comprehensive financial analysis reports without manual modeling or query engineering. MMRepAgent integrates multi-source data through a modular financial toolbox consisting of four key components: a Stock Factorization Tool for structured market indicators, a News Factorization Tool for semantic event modeling via SRL and SDPG, a Return Forecasting Tool based on an extended Pama-French model incorporating news impact and temporal effects, and a Risk Assessment Tool that enhances EGARCH-based volatility modeling with event-driven dynamics. The outputs from these modules are synthesized by a large language model to produce a multi-modal report containing textual insights, predictive visualizations, and event knowledge graphs. Extensive experiments on real-world datasets demonstrate the effectiveness, interpretability, and usability of MMRepAgent, showing substantial improvements in return forecasting accuracy, risk estimation quality, and report clarity compared to competitive baselines.

1. Introduction

Financial decision-making has become increasingly data-driven, yet the gap between complex modeling techniques and the needs of ordinary investors remains substantial. While institutional investors rely on advanced analytics and real-time data fusion systems, most individual users still face considerable barriers in understanding the interplay between financial news and quantitative indicators. The proliferation of unstructured information, especially from financial news and disclosures, offers rich signals-but only *if users can interpret and integrate them effectively*.

Recent studies [1] have explored two primary directions to enhance stock prediction. On one hand, structured stock data (e.g., prices, volumes, factors) is analyzed via time-series modeling and deep learning techniques. On the other, unstructured news text is processed using natural language processing (NLP) to extract event-driven insights [2–4]. For instance, Cai et al. [5] leverage TF-IDF to relate keyword sentiment

with market indices, while Sawhney et al. [6] integrate social media and news into multi-source temporal fusion frameworks.

However, these approaches often assume users possess technical knowledge and domain expertise. As illustrated in Fig. 1(a), retail investors typically rely on manual reading and rudimentary tools to make decisions that are inefficient and error-prone. Although recent advances in large language models, such as ChatGPT and BloombergGPT, allow for interactive financial analysis [7], their effectiveness hinges on well-posed queries that reflect a deep understanding of finance (Fig. 1(b)).

To address this challenge, we propose **MMRepAgent**, a fully automated, multimodal system designed to bridge the gap between data complexity and user accessibility. As shown in Fig. 1(c), the user simply submits a ticker or news snippet, and MMRepAgent autonomously retrieves structured stock data and related news, performs a series of targeted analyses, and outputs a unified, human-readable report. The generated report integrates textual interpretations, predictive image

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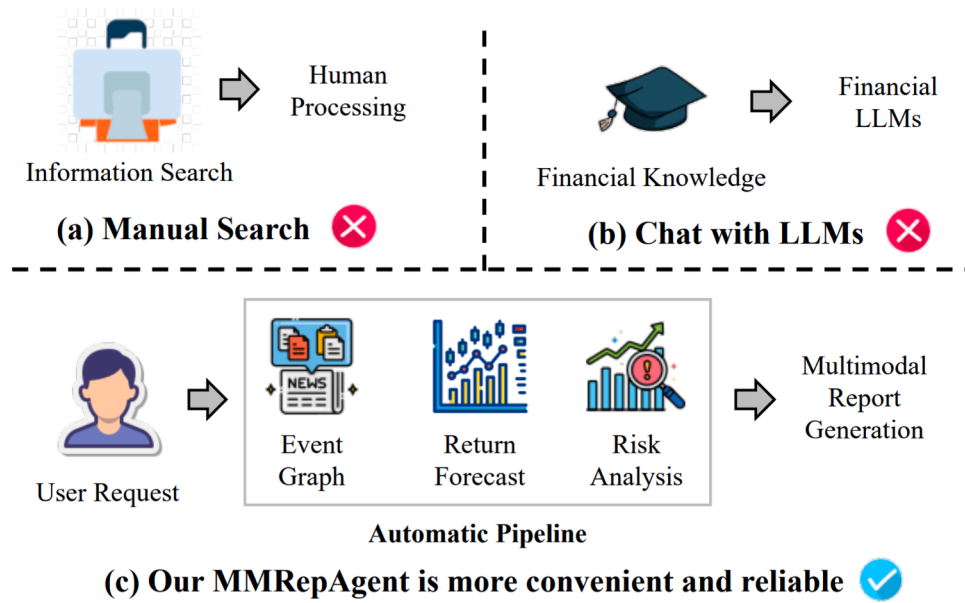


Fig. 1. Examples of the manual search, Chat with LLMs and our MMRepAgent solution.

visualizations, and knowledge graphs to support intuitive understanding and actionable insights. The overall framework is presented in Fig. 2.

Specifically, MMRepAgent features a modular architecture consisting of two main components: a *User Request Module*, which interprets natural language queries and maps them to structured analytical tasks, and a *Financial Toolbox*, which contains four domain-specific modules. The **Stock Factorization Tool** extracts and encodes quantitative market indicators across valuation, size, momentum, and liquidity dimensions. The **News Factorization Tool** transforms unstructured financial news into structured event graphs using semantic role labeling (SRL) and dependency parsing (SDPG), allowing temporal and causal event reasoning. The **Return Forecasting Tool** integrates these event-level news embeddings with stock factors into an enhanced Fama-French model, further augmented by time-periodic modulation to capture short-, mid-, and long-term effects. The **Risk Assessment Tool** quantifies volatility and downside risk through an event-aware extension of the EGARCH model and generates personalized Value-at-Risk (VaR) and Extreme Value-at-Risk (EVaR) estimates. These modules work in tandem to produce a unified analysis, which is then interpreted and structured by a large language model (LLM) into a multi-modal financial report.

Our key contributions are summarized as follows:

- We propose **MMRepAgent**, an end-to-end agent framework that automatically transforms user inputs into comprehensive multimodal financial reports by unifying structured data, news events, and predictive modeling.
- We develop an explainable and extensible *modular analysis pipeline*, which decomposes financial reasoning into four functional tools: factor extraction, event modeling, return prediction, and risk assessment.
- We conduct extensive experiments on real-world datasets, demonstrating that our system achieves superior forecasting accuracy, effective risk control, and high-quality report generation validated by domain experts.

2. Related work

2.1. Quantitative finance

Quantitative finance applies mathematical and statistical tools to model financial markets [8,9]. Core areas include *derivatives pricing* [10,11] and *risk analysis* [12], where models such as GARCH and

EGARCH [13] are widely used to capture time-varying volatility for forecasting and portfolio management [14]. In asset pricing, the Fama-French multifactor model extends CAPM by incorporating size, value, profitability, and investment factors, providing a foundation for both empirical studies and quantitative strategies [15]. To assess financial risk, Value at Risk (VaR) [16] remains a standard, with extensions like Conditional VaR and stress testing further addressing tail risk. Recently, researchers have explored integrating alternative data sources and exogenous signals to improve traditional models [17], though structured financial indicators remain the primary focus, limiting the modeling of unstructured, event-driven information [18].

2.2. News semantic understanding

Extracting structured semantics from financial news is essential for enabling downstream quantitative analysis. Among core techniques, *Semantic Role Labeling* (SRL) [19] identifies key predicates and their associated arguments (e.g., agents, patients), providing a high-level abstraction of event structures [20]. Complementarily, *Semantic Dependency Parsing Graphs* (SDPGs) [21] represent sentences as directed graphs, capturing fine-grained inter-word relations beyond surface syntax [22]. Standardized benchmarks [23] have driven the development of SDPG parsers, with methods ranging from graph-based neural models [24–26] to optimization-based decoders such as integer linear programming [27] and novel shift-reduce automata [28,29]. These tools offer structured representations that support complex reasoning tasks [30]. In financial domains, these semantic tools are increasingly used to transform unstructured news into graph-based or role-labeled formats, enabling downstream models to extract interpretable event-level insights and enhance market prediction or risk analysis pipelines [31].

2.3. LLM-Based agents and multimodal analysis

Large Language Models (LLMs) have made significant strides in the field of Natural Language Processing (NLP) in recent years, enabling a wide range of applications powered by artificial intelligence [32–34]. When combined with autonomous agents [35], which can interact with these LLMs and execute decisions based on their outputs, the potential applications expand even further. Concurrently, multimodal analysis integrates data from different modalities such as text, images -to provide a more comprehensive understanding of complex information [36]. In the

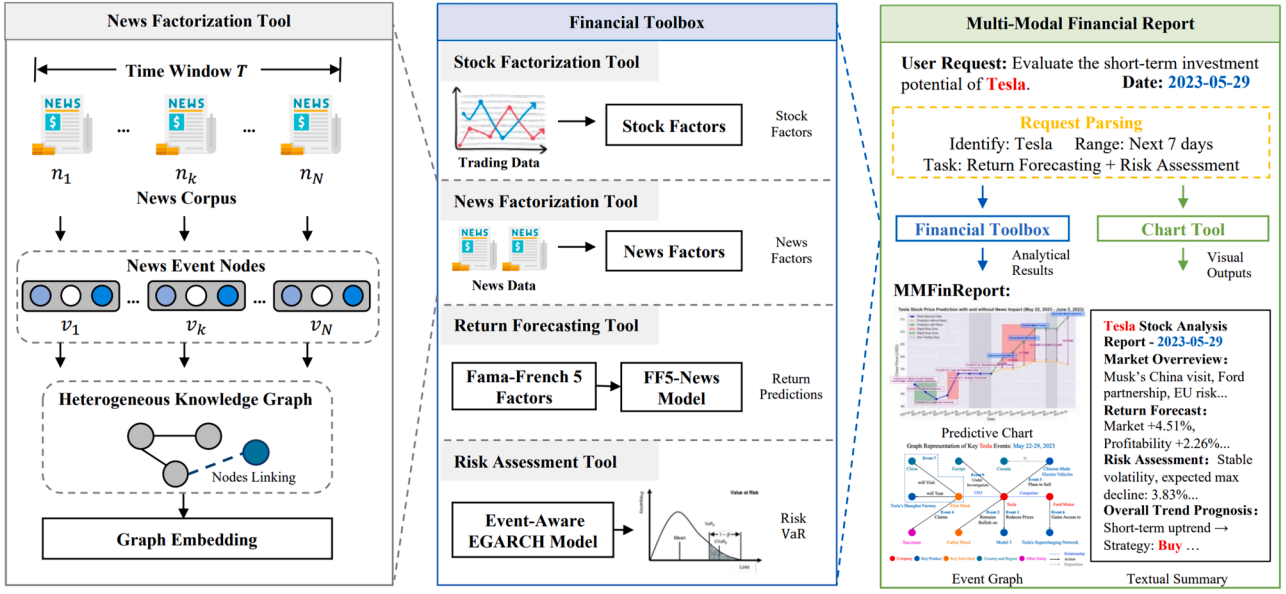


Fig. 2. The overall architecture of our MMRepAgent for explainable stock returns forecasting.

financial sector, the combination of LLMs and multimodal analysis offers a wide range of potential applications [37]. For instance, investment firms can utilize these technologies to analyze textual information from company earnings reports while simultaneously integrating stock market charts and social media sentiment analysis to assess market trends and investment risks [38]. Specifically, LLMs can automate the processing of analyst reports, news articles, and investor feedback to extract key insights and generate market analyses [39]. Furthermore, using image recognition technology, financial institutions can analyze product images, logos, and visual content related to brands, thereby conducting more comprehensive market research.

3. Proposed method

In this section, we introduce **MMRepAgent**, a user-driven framework for generating interpretable, multi-modal financial reports based on heterogeneous data sources. The system is designed to support individual investors by automating the process of information collection, factor-based analysis, and report synthesis.

MMRepAgent addresses three major challenges in financial analytics: (1) understanding the impact of complex and diverse news events on market behavior, (2) modeling the temporal dynamics and multi-scale patterns in stock movements, and (3) producing integrated outputs that combine predictive signals with human-readable insights.

3.1. Overall workflow of MMRepagent

As illustrated in Fig. 2, **MMRepAgent** is designed as a task-driven intelligent agent that autonomously executes financial reasoning workflows based on user requests. The system follows a structured three-stage process:

(1) *User Request as Perception*: The agent begins by interpreting the user query q_t , determining the analytical objective (e.g., return forecasting or risk assessment), and retrieving the corresponding financial and news context. This serves as the perception module, conditioning the downstream reasoning on personalized investment intent.

(2) *Toolbox-Based Reasoning*: The core analytical process is handled by the modular **Financial Toolbox**, which comprises four specialized tools: Stock Factorization, News Factorization, Return Forecasting, and Risk Assessment. These tools are selectively invoked based on the task,

enabling reusable, composable reasoning pathways that adapt to diverse financial scenarios.

(3) *Report as Action*: The final stage synthesizes results into a *Multi-modal Financial Report*, integrating structured textual narratives, predictive charts, and event knowledge graphs. This output bridges complex data modeling with interpretable insight, completing the agent's sense-reason-act loop.

In essence, MMRepAgent functions as a task-aware financial assistant, offering interpretable, personalized, and scalable financial intelligence for real-world decision-making.

3.2. Preliminaries

We formalize the financial report generation task as a user-driven modeling problem over temporal financial data. Let $\mathcal{T} = \{1, 2, \dots, T\}$ denote the trading timeline. At each time $t \in \mathcal{T}$, the system receives a user request q_t , a set of financial news events $\mathcal{N}_t = \{n_t^{(1)}, n_t^{(2)}, \dots\}$, and a structured market feature vector $s_t \in \mathbb{R}^{d_s}$. To capture temporal dependencies, we introduce a multi-scale window $\mathcal{W}_t$ centered at t , which incorporates recent short-term and longer historical context.

Given the input sequence $\{(q_t, \mathcal{N}_t, s_t, \mathcal{W}_t)\}_{t=1}^T$, the goal is to generate a corresponding output sequence $\{\mathcal{O}_t\}_{t=1}^T$, where each $\mathcal{O}_t$ denotes a multi-modal financial report. Each report includes three key outputs: a return prediction $\mathcal{R}_t \in \mathbb{R}$, a risk estimate $\mathcal{Q}_t \in \mathbb{R}$, and an interpretable report that integrates analytical texts, event graphs, and predictive charts. Formally, we define the system as a mapping function:

$$f_\theta : \{(q_t, \mathcal{N}_t, s_t, \mathcal{W}_t)\}_{t=1}^T \rightarrow \{\mathcal{O}_t\}_{t=1}^T, \quad (1)$$

This formulation establishes the unified input-output interface for MMRepAgent, bridging user-driven intent, multi-modal market signals, and financial reporting.

3.3. User request

The *User Request* module serves as the entry point of MMRepAgent, enabling the system to operate in a task-driven rather than data-driven manner. Each request, denoted as q_t , represents an investor's specific analytical intent at time t , such as querying the outlook of a particular stock or assessing portfolio risk under recent news events. This module parses the request to determine the relevant targets, filtering and aligning the appropriate financial data and news context for downstream analysis.

By conditioning the entire analytical pipeline on user input, the system supports personalized financial reasoning, allowing for report generation that is focused, interpretable, and decision-oriented.

3.4. Financial toolbox

3.4.1. Stock factorization tool

The Stock Factorization Tool processes structured market data to produce a quantitative representation of stock-specific characteristics across a target universe. At each time step t , for a given set of N stocks and m predefined financial factors (e.g., valuation, momentum, volatility), the tool constructs a cross-sectional factor matrix:

$$\mathbf{X}_f^{(t)} = \begin{bmatrix} f_1^1 & \dots & f_1^N \\ f_2^1 & \dots & f_2^N \\ \vdots & \ddots & \vdots \\ f_m^1 & \dots & f_m^N \end{bmatrix}, \quad (2)$$

where f_m^i denotes the m -th factor value of the i -th stock at time t .

This matrix $\mathbf{X}_f^{(t)}$ encodes factor signals in a structured format and serves as the input to downstream modules such as return forecasting and risk assessment. The factor set is updated dynamically over time to reflect recent market conditions.

3.4.2. News factorization tool

The News Factorization Tool transforms unstructured financial news into structured representations aligned with stock-level predictive tasks. Given a set of news articles $\mathcal{N}_t = \{n_t^{(1)}, n_t^{(2)}, \dots\}$ at time t , we extract event-level semantic information by combining outputs from pretrained encoders (RoBERTa), semantic role labeling (SRL), and semantic dependency parsing (SDPG). Each news item n_i is encoded as:

$$v_i = [e_{\text{SRL}}; e_{\text{SDPG}}; e_{\text{temporal}}], \quad (3)$$

where e_{SRL} and e_{SDPG} represent structured linguistic roles and dependencies, and e_{temporal} denotes the temporal encoding of publication time t_i .

To model event-level interactions, we construct a heterogeneous news-event knowledge graph $G = (V, E, A)$, where $V = \{v_i\}$, E encodes semantic, temporal, and impact-based edges, and A represents multi-type adjacency tensors. Node features are aggregated via attention-enhanced graph neural networks:

$$\mathbf{H}_n = \sum_i \alpha_i \cdot h_i \cdot D(\Delta t), \quad (4)$$

where h_i is the attention-pooled vector for event v_i , α_i is a temporal softmax score, and $D(\Delta t) = \exp(-\lambda \Delta t)$ is a decay function modeling recency effects (Fig. 3).

To incorporate market-level factor information, we align the news graph embedding $\mathbf{H}_n$ with the previously defined stock factor matrix $\mathbf{X}_f$, and concatenate them into a combined feature representation:

$$\mathbf{X} = \mathbf{W}_\alpha \odot [\mathbf{H}_n; \mathbf{X}_f], \quad (5)$$

where $\mathbf{W}_\alpha$ is a learnable weight matrix. The combined features are passed through a two-layer MLP with softmax output:

$$\mathbf{y}_n = \text{Softmax}(\text{MLP}(\mathbf{X}_{[:,n]})), \quad (6)$$

The definition of classification $\mathbf{y}_n$ in the Zou et al. [40] is followed, where mainly divided into three categories: positive, neutral, and negative:

$$\mathbf{E} = \begin{cases} \text{positive} & \text{if } r \in \text{top } 20\%, \\ \text{neutral} & \text{if } r \in 20\% \sim 60\%, \\ \text{negative} & \text{otherwise.} \end{cases} \quad (7)$$

The module is trained with cross-entropy loss:

$$\mathcal{L}_{nf} = \text{CrossEntropy}(\mathbf{y}, \mathbf{E}), \quad (8)$$

This allows the system to derive event-aware, stock-specific return predictions in response to real-time news dynamics.

3.4.3. Return forecasting tool

The Return Forecasting Tool enhances traditional financial models by incorporating dynamic, news-driven signals into the Fama-French 5-factor framework. This integration allows the system to capture both structural factor exposures and time-sensitive event effects in stock return prediction.

Formally, we extend the standard FF5 model by introducing news impact factors modulated by time-dependent periodic functions:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + b_i S M B_t + h_i H M L_t + r_i R M W_t + c_i C M A_t + \sum_k \left(w_k M_{k,t} \cdot P(t) \right), \quad (9)$$

where R_{it} is the return of stock i , R_{ft} the risk-free rate, and R_{mt} the market return at time t . The terms $\beta_i, b_i, h_i, r_i, c_i$ denote stock-specific exposures to classical risk factors. $M_{k,t}$ is the k -th news impact factor derived from the News Factorization Tool, and $P(t)$ models the temporal decay or persistence of that factor.

To account for different temporal behaviors of news, we define $P(t)$ as a linear combination of short-, mid-, and long-term basis functions:

$$P(t) = w_s \exp(-\lambda_s t) + w_m \sin(\omega t) + w_l \tanh(\alpha t), \quad (10)$$

where w_s, w_m, w_l are learnable weights controlling the relative contributions of immediate reaction, cyclical fluctuation, and long-term stabilization.

The model is trained by minimizing the mean squared error between the predicted return $\hat{R}_{it}$ and the observed return R_{it} :

$$\mathcal{L}_{\text{forecast}} = \sum_{i,t} (R_{it} - \hat{R}_{it})^2, \quad (11)$$

By embedding time-aware news effects into a factor-based framework, this module improves the granularity and interpretability of return forecasts, and enhances the responsiveness of the system to real-world events.

3.4.4. Risk assessment tool

The Risk Assessment Tool estimates market uncertainty by extending the classic EGARCH model to account for real-time news effects and stress conditions. This enables dynamic, event-aware volatility modeling and tail risk quantification.

To incorporate event-driven effects into volatility estimation, we introduce an enhanced conditional variance equation:

$$\ln \sigma_t^2 = \omega + \sum_{l=1}^p \alpha_l (|e_{t-l}| - \mathbb{E}|e_{t-l}|) + \sum_{j=1}^q \beta_j \ln \sigma_{t-j}^2 + \sum_{k=1}^q \gamma_k e_{t-k}^2 + \sum_{m=1}^M \delta_m I_m(t), \quad (12)$$

where σ_t^2 is the conditional variance at time t , e_t denotes residual shocks, and $I_m(t)$ captures the influence of the m -th event. The coefficients δ_m quantify event-specific impacts on volatility dynamics.

Based on this formulation, the volatility-aware Value at Risk (VaR) is computed as:

$$\text{VaR}_t = \mu_t - \sigma_t Z_\alpha + \eta \sum_{i=1}^N w_i I_i(t), \quad (13)$$

where μ_t is the expected return, Z_α is the quantile at confidence level α , and the term $\sum w_i I_i(t)$ adjusts risk according to news-derived shocks.

To simulate adverse conditions, we define a scenario set $S = \{s_1, s_2, \dots, s_k\}$, where each s_i aggregates shocks from multiple events. The stress-adjusted VaR is given by:

$$\text{VaR}_{\text{stress}} = \text{VaR}_{\text{base}} + \sum_{i=1}^k \text{Impact}(s_i), \quad (14)$$

Finally, we quantify tail risk using Expected Shortfall (ES):

$$\text{EVar} = \mathbb{E}[\text{Loss} \mid \text{Loss} > \text{VaR}], \quad (15)$$

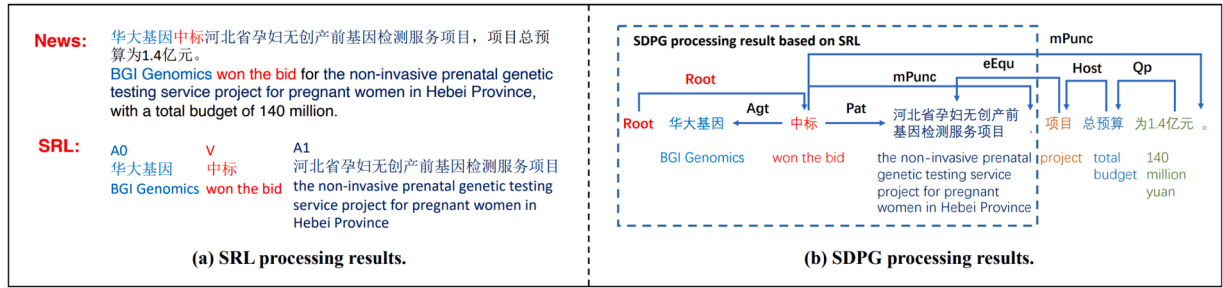


Fig. 3. The processing results of SRL and SDPG.

which reflects the average loss when market outcomes exceed the VaR threshold.

This module enriches conventional volatility modeling by integrating event timing and severity, offering a comprehensive framework for proactive risk evaluation in uncertain markets.

3.5. Multi-modal report generation

The final stage of our MMRepAgent synthesizes a human-readable, multi-modal financial report that integrates quantitative analysis and interpretability. This report combines textual summaries, predictive visualizations, and event-centric knowledge graphs to provide stakeholders with actionable insights.

Formally, the generation process is controlled by a large language model (LLM), which takes three key inputs: stock return predictions $\mathcal{R}$, risk estimates VaR, and a structured prompt $\mathcal{M}$ defining the desired output format and content. The LLM outputs a unified report:

$$\mathcal{O} = \text{LLM}(\mathcal{R}, \text{VaR}, \mathcal{M}) = \{\text{Text_Report}, \text{Charts}, \text{Knowledge_Graph}\}. \quad (16)$$

The prompt $\mathcal{M}$ is designed to guide the LLM in assembling different analytical outputs into a coherent structure. It encodes high-level instructions such as: (1) presenting return predictions across multiple factor dimensions (e.g., market, size, valuation, profitability, investment, and news effect), (2) visualizing risk metrics including VaR dynamics, (3) constructing trend charts for price and volatility trajectories, (4) generating a knowledge graph linking news events to return impacts, and (5) summarizing findings into an investment recommendation.

By aggregating multimodal information into a unified report, this module bridges model outputs and user understanding, enhancing both explainability and decision support. The LLM-driven generation ensures flexibility across different user intents, while the structured content guarantees consistency in financial interpretation. More visualization of multimodal financial reports will be displayed in experiments.

4. Experiments

4.1. Dataset

The Financial News and Stock Price Integration Dataset (FNSPID)¹, proposed by [41], serves as the foundational dataset for this study. Covering a comprehensive period from January 1, 1999, to December 31, 2023, FNSPID encapsulates stock price and news data for 4775 S&P 500 companies over 9125 days. This extensive dataset includes approximately 29.7 million stock price records and 15.7 million time-aligned financial news entries, making it a robust resource for time-series financial analysis and predictive modeling.

Specifically, FNSPID comprises daily stock price data sourced from Yahoo Finance, featuring key metrics such as opening price (P_{open}), highest price (P_{high}), lowest price (P_{low}), closing price (P_{close}), adjusted

closing price (P_{adj}), and trading volume (V). Complementing these quantitative records, the dataset incorporates financial news articles collected from reputable sources, including NASDAQ, Bloomberg, Reuters, and Benzinga. Each news entry provides a timestamp (T_{news}), headline, and full article text, enabling detailed investigations into market sentiment and event-driven price dynamics. As a distinctive feature, FNSPID integrates both stock prices and news data, offering a dual-modality perspective critical for advanced financial forecasting.

To facilitate model training and evaluation, we partition FNSPID into three distinct subsets following a chronological strategy. The training set spans from January 1, 1999, to December 31, 2018, encompassing 20 years of historical data to capture long-term market patterns. The validation set extends from January 1, 2019, to December 31, 2020, providing a two-year window to fine-tune model parameters. Finally, the test set covers January 1, 2021, to December 31, 2023, allowing assessment of predictive performance on recent market conditions over three years. This temporal division ensures that our models leverage extensive historical trends while maintaining relevance to contemporary financial scenarios, aligning with the sequential nature of time-series prediction tasks.

4.2. Evaluation metrics

Firstly, we utilized accuracy, F1 score, recall, and precision as evaluation metrics for our news factorization module, same as [40]. Secondly, we employed the GRS (Goodness-of-Fit R-squared) [42] metric, its p-value and the mean absolute value of alpha to measure the model's explanatory power on return predictions for explainable news effects. For VaR (Value at Risk) risk assessment, we used RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and VaR Loss Coverage Rate as evaluation metrics. In assessing multimodal report generation, we rely on an expert rating score, where domain experts evaluate the reports' readability and practical utility on a standardized scale. This qualitative metric ensures that the generated reports meet the needs of end users, bridging analytical outputs with real-world applicability. For trading performance evaluation, we conduct simulated backtesting and adopt six widely-used financial indicators: Total Return (%), Compound Annual Growth Rate (CAGR, %), Sharpe Ratio, Calmar Ratio, Maximum Drawdown (%), and Win Rate (%). These metrics collectively measure profitability, risk-adjusted performance, downside risk, and trading consistency in real-world scenarios.

4.3. Implementation details

To ensure reproducibility and consistency, we adopt a unified hyperparameter search grid. The learning rate is tuned within $[1 \times 10^{-3}, 1 \times 10^{-4}, 1 \times 10^{-5}, 1 \times 10^{-6}]$, and the batch size ranges from [16, 32, 64, 128]. A dropout rate of 0.1 is applied to mitigate overfitting, while the MLP in the news factorization module uses a hidden size of 1024 units to capture complex financial features. For knowledge graph construction, we introduce a pre-trained RoBERTa as the text encoder with a learning rate of 5×10^{-5} and train it for 10 epochs to extract semantic information from

¹ https://github.com/Zdong104/FNSPID_Financial_News_Dataset

Table 1

Evaluating the News Factorization module via downstream stock return classification performance. Best results are highlighted in **bold**.

Model	Reference	Resource	Accuracy	F1 Score	Recall	Precision
StockNet [43]	ACL 2018	News	45.92	45.65	47.15	46.39
KDTCN [44]	WWW 2019	News	55.61	53.18	52.03	52.63
FactorVAE [45]	AAAI 2022	Factors	60.39	59.88	57.45	58.64
DeepTrader [46]	AAAI 2021	Factors	62.91	60.65	60.94	60.79
CausalStock [47]	NeurIPS 2024	News	64.26	62.83	64.81	63.78
StockFormer [48]	LJCAI 2023	Factors	66.88	65.41	64.54	64.97
Self-supervised SRLP [40]	FinNLP 2022	News	63.21	61.62	61.76	61.69
		News + Factors	68.32	66.29	64.01	65.13
MDGNN [49]	AAAI 2024	News	63.69	64.55	64.32	64.43
		News + Factors	70.82	68.53	68.12	68.32
		Factors	62.65	63.47	62.64	63.05
FinReport [39]	WWW 2024	News	65.83	66.41	63.23	64.78
		News + Factors	75.58	75.76	72.16	73.92
		Factors	67.99	67.42	65.76	66.58
FinAgent [50]	KDD 2024	News	65.05	63.41	64.17	63.79
		News + Factors	76.09	76.52	74.22	75.35
		Factors	70.75	69.92	70.28	70.11
MMRepAgent	Ours	News	72.28	70.55	72.42	71.46
		News + Factors	79.72	78.84	78.61	78.72

financial news. This setup effectively captures key event details, such as corporate actions or policy shifts, laying a robust foundation for graph-based modeling. The resulting features are fed into a 2-layer graph convolutional network (GCN) with a hidden size of 512, optimizing the graph representation by modeling relational dependencies (e.g., causal or temporal links) in financial events. This enhances the system's ability to represent and predict intricate financial dynamics. For the report generation module described in Section 3.5, we employ GPT-4-Turbo version gpt-4-turbo-2024-04-09, selected for its instruction-following capabilities in synthesizing multimodal content.

All experiments are conducted on eight NVIDIA 4090 GPUs, with models optimized via the AdamW optimizer. This setup ensures computational efficiency and consistency, supporting the scalability of our financial event modeling framework. In terms of model size, MMRepAgent contains approximately 152M trainable parameters, making it comparable to BERT-base and suitable for deployment after lightweight quantization. We further measure inference efficiency over 50 representative queries. On eight NVIDIA 4090 GPUs, the average time-to-first-token is 0.12s and the total generation time for an analytical report (about 1200 tokens) is 1.04s. On a mobile device simulator (Snapdragon 8 Gen 2, 8GB RAM) with INT8 quantization, the time-to-first-token increases to 0.41s and the average report generation time is 3.26s. These results demonstrate that MMRepAgent maintains acceptable responsiveness and is computationally feasible for retail investors even on mobile platforms.

4.4. News factorization evaluation

To evaluate the effectiveness of our enhanced news factorization module and the benefit of integrating multi-source information, we compare our method with a comprehensive set of baseline models. As shown in Table 1, we categorize these baselines according to their input modalities: 1) News-only models, including StockNet [43], KDTCN [44] and CausalStock [47]. 2) Factor-only models, including FactorVAE [45], DeepTrader [46] and StockFormer [48]. 3) methods are integrated both news and factors information, including Self-supervised SRLP [40], MDGNN [49], FinReport [39] and FinAgent [50].

The experimental results are presented in Table 1. The following observations can be made: 1) Models that integrate both news and factor information consistently outperform those relying on a single source. For instance, MDGNN improves from 63.69% (news-only) and 62.65%

(factor-only) to 70.82% when combining both. Similarly, FinReport increases from 65.83% (news-only) and 67.99% (factor-only) to 75.58% with both sources. This trend confirms that financial market behaviors are influenced by both structured financial indicators and unstructured textual events, and neither can be ignored for effective prediction. 2) For the same model architecture, the joint use of news and factors yields consistent performance gains over their single-modality variants. This observation is valid across different models such as SRLP, FinReport, and FinAgent. For example, FinAgent achieves 76.09% accuracy in the joint setting, compared to 65.05% and 70.75% in the news-only and factor-only cases, respectively. These improvements verify that the two modalities contain complementary signals that reinforce model decision-making. 3) Our model MMRepAgent surpasses all other baselines across all metrics. It achieves 79.72% accuracy, 78.84% F1 score, 78.61% recall, and 78.72% precision. We attribute this superior performance to several innovations: i) the incorporation of multiple semantically rich news events instead of single headlines; ii) the construction of an event-level knowledge graph capturing temporal, semantic, and causal relations; and iii) the deployment of a large language model to abstract, filter, and align high-level financial semantics. In addition, even when restricted to news-only or factor-only inputs, MMRepAgent still achieves top-tier performance (72.28% and 70.55% accuracy respectively), outperforming all other models in the same category. This indicates that our news factorization and representation mechanisms are inherently more expressive and robust, offering substantial predictive power even without auxiliary modalities. These findings demonstrate the effectiveness and versatility of our factorization architecture, reinforcing the value of joint modeling and advanced semantic abstraction in real-world financial forecasting.

4.5. Return forecasting research

To rigorously evaluate the explanatory contribution of the proposed News Effect Factor, we employ the GRS test (Gibbons, Ross and Shanken) [51] and complement it with direct statistical tests on asset-level pricing errors. We first recall the standard definition of the GRS statistic. Let T denote the number of time observations, N the number of test assets, $\hat{\alpha}$ the $N \times 1$ vector of estimated pricing errors (alphas), $\hat{\Sigma}$ the $N \times N$ sample covariance matrix of residuals, $\hat{f}$ the $K \times 1$ vector of sample mean factor returns, and $\hat{\Sigma}_f$ the $K \times K$ sample covariance of

Table 2

Fama-French 5 factors with news effect factor model GRS test results, paired t -test and Wilcoxon signed-rank results, and VaR risk assessment system evaluation results.

Model	GRS	GRS p-value	Mean Absolute Value of Alpha
Fama-French 5 Factors	3.6721	2.135×10^{-3}	0.0728
Fama-French 5 Factors + News Effect Factor	6.0295	1.205×10^{-5}	0.0554
Paired t -test (FF5 vs FF5-News)	$\bar{d} = 0.0174$, $s_d = 0.06$, $N = 420$ $t = 5.94$, $df = 419$, $p = 4.10 \times 10^{-8}$		
Wilcoxon signed-rank	two-sided $p = 2.3 \times 10^{-6}$ (robustness)		
Model	RMSE	MAE	VaR Loss Coverage Rate
VaR Risk Assessment System	0.0873	0.7935	0.8451

factor returns. The GRS statistic is defined as:

$$GRS = \frac{(T - N - K)}{N} \cdot \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{f}' \hat{\Sigma}_f^{-1} \hat{f}}. \quad (17)$$

The GRS is distributed as an F -statistic under the null hypothesis that all elements of $\hat{\alpha}$ are zero simultaneously. Importantly, a larger GRS provides stronger evidence against the null hypothesis (i.e., against the joint zero-alpha hypothesis) and therefore signals greater model misspecification rather than model superiority. Table 2 reports both the GRS results and a direct statistical assessment of the reduction in mean absolute pricing errors. As shown, the baseline Fama-French 5-Factor model (FF5) and the FF5 model augmented with the News Effect Factor (FF5-News) yield the following summary statistics on our OOD sample (2021-01-01 to 2023-12-31): the mean absolute alpha decreases from 0.0728 (FF5) to 0.0554 (FF5-News). At the same time, the GRS statistic increases from 3.6721 to 6.0295 with correspondingly smaller p -values. Because the GRS depends not only on the vector of alphas but also on the residual covariance matrix $\hat{\Sigma}$ and on $\hat{f}' \hat{\Sigma}_f^{-1} \hat{f}$, it is possible for the GRS value to rise even when the cross-sectional mean magnitude of alpha decreases. Such a phenomenon can occur when the residual cross-sectional covariance or the factor-sample-moments change between model specifications. Therefore, we caution against interpreting the raw GRS number in isolation and focus instead on economically meaningful reductions in pricing errors and their statistical significance. To formally test whether the observed reduction in mean absolute alpha is statistically significant, we perform a paired t -test on asset-level absolute alphas. For each asset i we compute

$$d_i = |\hat{\alpha}_i^{FF5}| - |\hat{\alpha}_i^{FF5-News}|, \quad (18)$$

and evaluate the null hypothesis $H_0 : \mathbb{E}[d_i] = 0$ against a two-sided alternative. The paired t -statistic is computed as

$$t = \frac{\bar{d}}{s_d / \sqrt{N}}, \quad (19)$$

where $\bar{d} = \frac{1}{N} \sum_{i=1}^N d_i$ and s_d is the sample standard deviation of $\{d_i\}$. We report the degrees of freedom $df = N - 1$ and the two-sided p -value. As a robustness check we also compute the Wilcoxon signed-rank test (non-parametric) on $\{d_i\}$. In our OOD evaluation we select $N = 420$ assets with sufficient news coverage in the 2021-2023 period. The paired t -test yields $\bar{d} = 0.0174$, $s_d = 0.06$, $t = 5.94$ with $df = 419$, and a two-sided p -value of 4.10×10^{-8} . The Wilcoxon signed-rank test is also highly significant with a two-sided p -value of 2.3×10^{-6} . These results indicate that the reduction in mean absolute alpha from 0.0728 to 0.0554 is statistically significant and not likely due to random sampling variation. In summary, although both model specifications are formally rejected by the GRS test in our OOD sample, the FF5-News model achieves an economically meaningful and statistically significant reduction in pricing errors. We therefore present the reduction in alpha (together with the paired t -test and Wilcoxon robustness results) as the primary evidence that the News Effect Factor improves the model's explanatory power.

4.6. VaR Risk assessment

In this section, we evaluate the effectiveness of our VaR risk assessment module. The experimental results are summarized in Table 2. Specifically, our system achieves an RMSE of 0.0873 and an MAE of 0.7935, indicating a high level of predictive accuracy. In addition, the VaR loss coverage rate reaches 0.8451, demonstrating that the system effectively captures a substantial portion of realized losses. These results confirm that our VaR risk assessment module provides reliable support for risk evaluation and can serve as a valuable tool for financial risk management.

4.7. Multicollinearity diagnostics

To evaluate potential multicollinearity arising from the shared news data in deriving the News Effect Factor alongside return forecasting and risk assessment components, we implement diagnostic procedures within the extended Fama-French five-factor (FF5-News) model. This assessment is crucial for affirming that inter-factor correlations do not inflate coefficient variances or impair the interpretability of our knowledge-based MMRepAgent framework, wherein news is structured into graphs to facilitate advanced financial reasoning.

We begin by estimating pairwise Pearson correlations between the News Effect Factor and each FF5 factor-market excess return (MKT), size (SMB), value (HML), profitability (RMW), and investment (CMA)-across the 2010 to 2023 sample period. Subsequently, we derive Variance Inflation Factors (VIFs) for all factors to measure multicollinearity extent. For each factor j , the VIF is computed via an auxiliary ordinary least squares regression of factor F_j against the remaining factors, yielding

$$VIF_j = \frac{1}{1 - R_j^2}, \quad (20)$$

where R_j^2 is the regression's coefficient of determination, VIF values above 5 suggest moderate multicollinearity, while those exceeding 10 indicate severe concerns requiring remediation.

Results reveal low pairwise correlations, ranging from 0.38 to 0.61, as detailed in Table 4. Correspondingly, VIF values for all factors are below 3, with the News Effect Factor at 2.1 and FF5 factors between 1.8 and 2.4. These findings demonstrate minimal multicollinearity, implying that the news-derived factor introduces orthogonal information without inflating estimate variances. This bolsters the reliability of our agent's factor attributions, ensuring that the knowledge graph integration enhances predictive accuracy without methodological artifacts in the multimodal financial analysis.

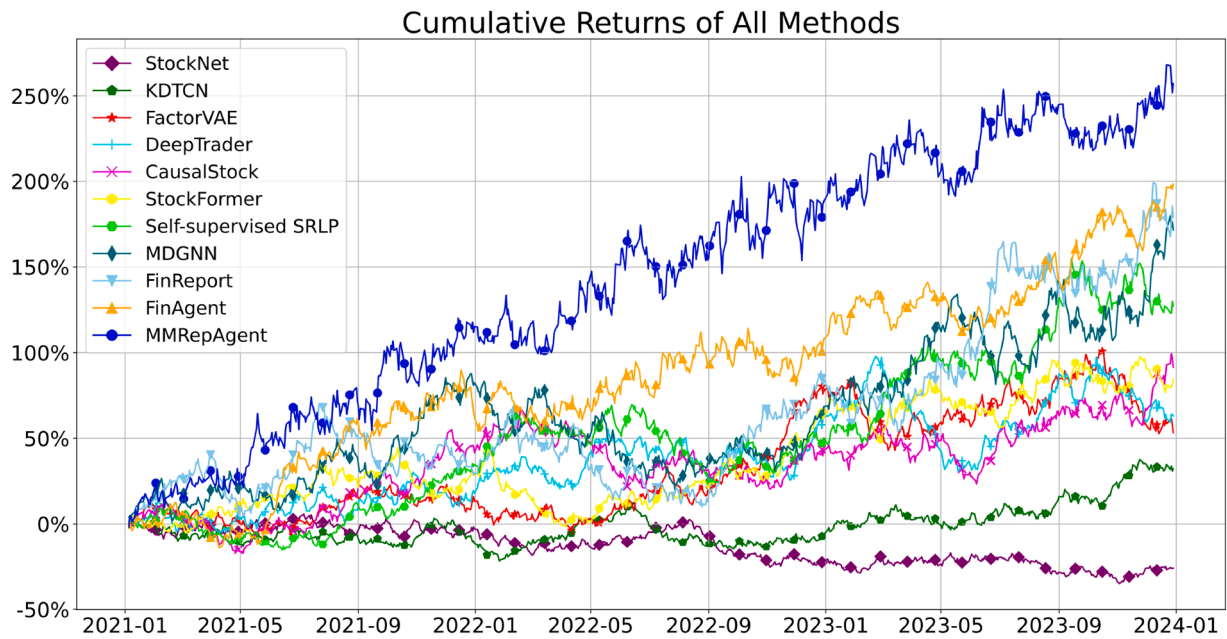
4.8. Profitability evaluation in real-world scenarios

To evaluate the real-world investment performance of our system, we conduct a backtest on the FNSPID dataset from January 1, 2021 to December 31, 2023. The trading strategy assumes buying at the market open and selling at the market close each day, which aligns with standard intraday operations in the U.S. equity market. Each day, our system

Table 3

Performance comparison of all baselines on 9 evaluation metrics. **Bold** represents optimal performance, while underline represents suboptimal.

Model	Total Return (%)	CAGR (%)	Sharpe Ratio	Calmar Ratio	Max Drawdown (%)	Win Rate (%)
StockNet	-24.64	-9.04	-0.46	-0.23	-38.55	35.07
KDTCN	35	10.58	0.39	0.43	-24.33	42.13
FactorVAE	51.07	14.83	0.55	0.61	-24.4	48.93
DeepTrader	59.7	16.98	0.62	0.75	-33.99	51.99
CausalStock	91.02	24.22	0.82	1.17	-27.88	50.06
StockFormer	79.33	21.62	0.76	0.88	-31.9	48.63
Self-supervised SRLP	123.59	30.95	1.03	1.35	-26.81	49.93
MDGNN	161.43	37.99	1.15	1.15	-36.21	51.26
FinReport	176.61	40.63	1.08	1.56	-35.12	<u>53.85</u>
FinAgent	<u>200.46</u>	<u>44.58</u>	<u>1.43</u>	<u>2.39</u>	<u>-23.65</u>	52.63
Our MMRepAgent	260.71	53.71	1.72	3.12	-16.21	58.76
Improvement (%)	(+30.06)	(+20.48)	(+20.28)	(+30.54)	(+31.46)	(+9.12)

**Fig. 4.** Comparing the performance of different models in real-world scenarios.**Table 4**

Pairwise correlations and VIF values for news effect factor and FF5 factors.

	MKT	SMB	HML	RMW	CMA	News Effect Factor
Pairwise Correlations (News Effect Factor)	0.41	0.38	0.52	0.44	0.61	-
VIF Values	1.82	1.96	2.33	2.07	2.14	2.18

identifies stocks with financial news, predicts their expected returns and risks, and selects the top-10 candidates to construct a portfolio. The portfolio must satisfy a VaR constraint (5% threshold), and a transaction fee of 0.1% is applied.

As shown in Table 3, our model MMRepAgent achieves the best overall performance across all key investment metrics. It records a total return of 260.71% and an annualized return (CAGR) of 53.71%, significantly surpassing the strongest baseline (FinAgent, 200.46% total return, 44.58% CAGR). This demonstrates the superior long-term profitability of our system. In terms of risk-adjusted returns, MMRepAgent obtains a Sharpe ratio of 1.72 and a Calmar ratio of 3.12, indicating that it not only delivers high returns but also maintains strong performance relative to its volatility and drawdowns. Compared with FinAgent (Sharpe: 1.43, Calmar: 2.39), our system improves by 20.28% and 30.54% respectively, reflecting more efficient return generation per unit of risk. Importantly, our method achieves the lowest maximum drawdown at -16.21%, which is 31.46% lower than the suboptimal base-

line (FinAgent: -23.65%). This underscores the system's ability to mitigate extreme downside risk, a critical feature for institutional-grade deployment. Finally, the win rate of 58.76% is the highest among all models, showing that our system maintains superior trading consistency across thousands of transactions. This stability is particularly valuable in volatile environments, where short-term losses can easily erode long-term gains.

We attribute these comprehensive improvements to three key factors: (1) a fine-grained multi-news event understanding that captures subtle market signals, (2) the integration of structured financial factors to complement unstructured text, and (3) a risk-aware portfolio construction process guided by dynamically adjusted VaR constraints. The synergy among these components enables the system to balance aggressiveness and caution, yielding high returns with disciplined risk control. Fig. 4 illustrates the cumulative return curves, where MMRepAgent consistently outperforms all baselines, further validating its robustness across varied market conditions.

Table 5

Ablation studies over different components. ✓ indicates the component is added to MMRepAgent. In the table, SFT, NFT, RFT, and RAT represent Stock Factorization Tool, News Factorization Tool, Return Forecasting Tool, and Risk Assessment Tool, respectively. **Bold** represents optimal performance.

SFT	NFT	RFT	RAT	Total Return (%)	CAGR (%)	Sharpe Ratio	Calmar Ratio	Max Drawdown (%)	Win Rate (%)
			✓	43.06	12.75	0.53	0.62	-20.68	47.86
✓		✓		116.08	29.46	1.12	0.98	-30.11	50.26
	✓	✓		136.91	33.31	1.17	1.19	-27.97	51.88
✓	✓	✓		173.38	40.08	1.29	1.78	-22.56	52.85
✓		✓	✓	208.54	45.87	1.45	2.49	-18.46	56.59
	✓	✓	✓	221.54	47.9	1.54	2.84	-16.86	55.26
✓	✓	✓	✓	260.71	53.71	1.72	3.12	-16.21	58.76

Table 6

Expert evaluation of 30 multi-modal financial reports across five dimensions. Each report was rated by 5 financial experts.

Evaluation Dimension	Avg. Score	Positive Rate (%)
Readability	0.86	91.3
Factual Completeness	0.78	88.0
Interpretability	0.82	90.0
Practical Utility	0.84	89.3
Overall Impression	0.85	90.7

4.9. Ablation study on system components

To evaluate the contribution of each individual component in our system, we conduct a series of ablation studies under the same experimental configuration described in the *Profitability Evaluation in Real-world Scenarios* Section 4.8. Table 5 reports the results of selectively enabling modules within MMRepAgent, including the Stock Factorization Tool (SFT), News Factorization Tool (NFT), Return Forecasting Tool (RFT), and Risk Assessment Tool (RAT).

We first examine the effect of using only the Risk Assessment Tool (RAT). Without any forecasting capability, the system still achieves a reasonable Sharpe ratio of 0.53 and limits the maximum drawdown to -20.68%, indicating that RAT effectively reduces downside risk by filtering out highly volatile assets. This suggests that even in the absence of return signals, RAT provides a defensive mechanism that protects capital during adverse market conditions.

Next, we analyze the effectiveness of the Return Forecasting Tool (RFT). When RFT is enabled with either SFT or NFT, the system achieves notable improvements in profitability, with total returns of 116.08% and 136.91%, respectively. When both SFT and NFT are used together with RFT, performance further improves to 173.38% total return and a Sharpe ratio of 1.29. This progression confirms that RFT is the core driver of return generation, and that combining structured financial factors with news-based signals significantly enhances predictive power.

We then consider the impact of integrating RAT on top of an already predictive framework. For example, comparing the configuration with RFT + NFT (without RAT) and RFT + NFT + RAT, we observe that the Sharpe ratio increases from 1.17 to 1.54, and the maximum drawdown reduces from -27.97% to -16.86%. Similarly, when RAT is added to RFT + SFT, the Sharpe ratio increases from 1.12 to 1.45, and the Calmar ratio nearly doubles. These results demonstrate that RAT complements RFT by stabilizing returns, improving drawdown control, and increasing the consistency of the trading strategy.

Finally, when all four modules are jointly activated, the full MMRepAgent model achieves the best performance across all evaluation metrics, including a total return of 260.71%, an annualized return of 53.71%, a Sharpe ratio of 1.72, and a win rate of 58.76%. This highlights the strong synergy among forecasting, factor modeling, and risk

control. The results confirm that each component makes a distinct and essential contribution to the system's real-world effectiveness.

4.10. Cross-Dataset generalization evaluation

To test the generalization of MMRepAgent and address the need for broader coverage across diverse market conditions and data sources, we evaluate the framework on three additional international benchmarks: FTSE 100 (UK blue-chip stocks, sourced via FintHub API for news and financial data), CSI 300 (China mainland A-shares, sourced via Tushare API), and Nikkei 225 (Japan equities, sourced via Nikkei Telecom). This assessment is essential for retail investors engaging with global assets, verifying the model's robustness in heterogeneous news ecosystems, economic volatilities, and regulatory environments. We maintain the same baselines as in 4.4, categorized by input modalities (News-only, Factor-only, and joint News + Factors), and adopt identical experimental configurations. To facilitate training and evaluation, each dataset is partitioned chronologically: training set from 2015 to 2018, validation set from 2019 to 2020, and test set from 2021 to 2023, aligning with the sequential nature of financial time-series tasks. The experiments apply the same multimodal pipeline: news is converted to knowledge graphs, fused with financial factors via agent reasoning, and evaluated on downstream stock return classification. Performance metrics include accuracy, F1 score, recall, and precision, computed over balanced test sets.

Results in Table 7 demonstrate a consistent pattern across datasets, closely mirroring the trends observed in the FNSPID evaluation. Models integrating both news and factor information consistently outperform those relying on a single source, validating the value of multimodal fusion in diverse contexts. MMRepAgent exhibits superior capability by achieving the highest results in all markets and on all metrics, driven by its advanced knowledge graph that effectively abstracts semantic relations for precise predictions. The framework's strong generalization is evidenced by its high performance consistency across heterogeneous datasets, maintaining accuracy above 76% despite varying market structures. MMRepAgent's applicability to different markets is supported by its adaptive performance, fusing high metrics with market-specific dynamics: on FTSE 100, it handles mature market uncertainties like Brexit impacts through temporal event relations in its knowledge graph, achieving 78.92% accuracy and enhanced recall (77.81%); on CSI 300, it navigates emerging market policy noise by capturing prolonged event effects, yielding 76.45% accuracy with balanced F1 (75.68%); and on Nikkei 225, it integrates Asian-specific signals such as yen fluctuations via semantic alignment, delivering 77.61% accuracy and strong precision (76.64%). These insights affirm MMRepAgent's overall superiority and robust generalization across markets.

4.11. Evaluation of multi-modal financial report generation

In this section, we evaluate the effectiveness and usability of our automatically generated multi-modal financial reports. Each report is generated based on the outputs of our Return Forecasting and Risk

Table 7

Cross-dataset evaluation of stock return classification performance. Best results are in **bold**. Metrics are averaged over News + Factors settings unless specified.

Model	FTSE 100 (UK)				CSI 300 (China)				Nikkei 225 (Japan)			
	Acc.	F1	Rec.	Prec.	Acc.	F1	Rec.	Prec.	Acc.	F1	Rec.	Prec.
News-only Baselines												
StockNet	45.12	44.28	45.61	44.94	40.85	40.39	40.72	40.55	45.89	45.12	46.34	45.73
KDTCN	54.78	53.51	53.23	53.37	51.03	49.79	49.14	49.46	55.92	54.68	54.11	54.39
CausalStock	63.29	61.89	63.65	62.76	58.41	57.08	58.55	57.81	64.58	63.14	64.92	63.95
Factor-only Baselines												
FactorVAE	60.41	59.13	57.89	58.50	55.58	54.45	54.04	54.25	61.72	60.51	59.96	60.23
DeepTrader	62.78	61.29	61.42	61.35	57.61	56.47	56.30	56.38	64.09	62.78	62.29	62.53
StockFormer	66.08	64.68	64.29	64.48	60.92	59.58	59.33	59.45	67.41	66.01	65.64	65.83
News + Factors Baselines												
SRLP	68.06	66.49	65.36	65.91	62.68	61.27	60.45	60.86	69.39	67.81	66.99	67.40
MDGNN	70.58	68.34	67.92	68.13	65.05	63.41	63.06	63.24	71.91	70.13	69.76	69.95
FinReport	75.01	75.19	71.81	73.53	69.29	69.47	66.77	68.08	76.34	76.52	73.71	74.99
FinAgent	73.88	74.31	71.93	73.15	68.15	68.58	66.55	67.56	75.21	75.64	73.50	74.57
MMRepAgent (Ours)	78.92	78.14	77.81	77.97	76.45	75.68	75.32	75.50	77.61	76.83	76.45	76.64

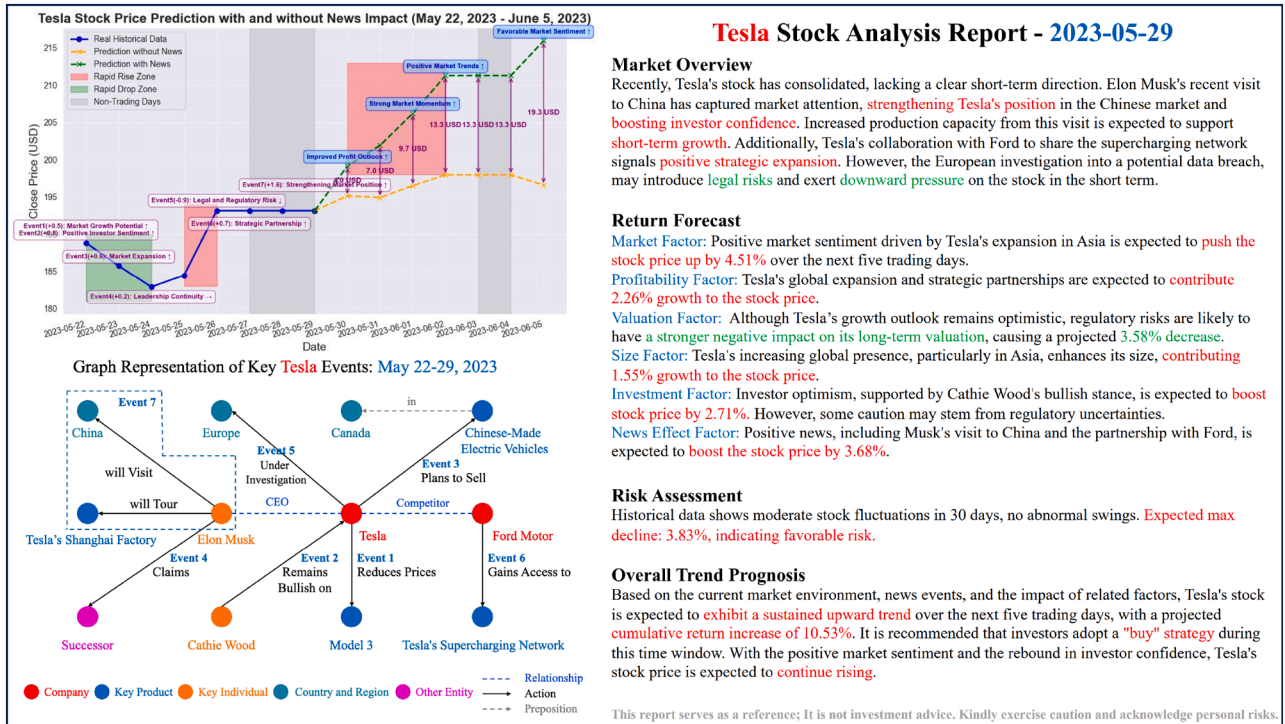


Fig. 5. Example of the multi-modal financial report generated by MMRepAgent.

Assessment modules, incorporating a wide range of factor dimensions such as market sentiment, profitability, valuation, size, and investment behavior, as well as the news effect factor. The reports are composed using a large language model (LLM), which transforms structured signals into coherent narrative explanations for end users. A representative example is shown in Fig. 5, illustrating our report on Tesla dated May 29, 2023.

The report integrates textual description, tabular forecasting results, and explicit investment suggestions. For example, in the Tesla report, the model synthesizes a positive market outlook based on Elon Musk's China visit, strategic cooperation with Ford, and supporting financial indicators. It concludes with a buy recommendation supported by a projected 10.53% five-day return and a low expected maximum loss of 3.83%, providing investors with both opportunity and risk assessments in a single document.

To evaluate the overall quality and user perception of our multi-modal financial reports, we conducted a structured expert review. We randomly selected 30 reports and invited five domain experts to score each report across five dimensions: readability, factual completeness, interpretability, practical utility, and overall impression. Each dimension was rated on a scale from -1 (poor), 0 (neutral), to 1 (good).

As shown in Table 6, the reports received consistently favorable feedback across all dimensions. Most notably, the *Overall Impression* score reached an average of 0.85 with a positive rate of 90.7%. These results indicate that the generated reports are not only fluent and well-structured but also factually grounded, interpretable, and practically useful for decision-making. Experts noted that the integration of structured forecasts, risk assessments, and narrative explanations provides a clear and actionable summary for investors, especially those without deep technical backgrounds.

4.12. Model limitations and failure case analysis

To provide a balanced evaluation of our framework's performance boundaries, we examine two representative cases from the 2021–2023 test period where predictions deviated substantially from realized outcomes under extreme market conditions. These cases involve rare tail-risk events characterized by systemic contagion or fundamental business model shifts, scenarios that are inherently challenging due to their limited representation in historical training data. By analyzing these cases, we aim to transparently communicate the operational scope of MMRepAgent and identify opportunities for future enhancement.

Our first case examines First Republic Bank during the March 2023 regional banking crisis. On March 13, 2023, following Silicon Valley Bank's collapse three days earlier, our system predicted a five trading days return of -38.51% for FRC based on negative news sentiment scores of 0.78 and financial sector factor deterioration. The prediction incorporated signals from 127 news articles published within 48 hours, indicating elevated market stress. However, the actual five-day cumulative return was -75.49% , representing a prediction error of -36.98 percentage points. This deviation reflects the challenge of modeling inter-bank contagion effects that exhibited nonlinear acceleration beyond historical patterns in our training data. The deposit flight and asset liquidation dynamics during this crisis significantly exceeded the volatility ranges observed in the post-2008 regulatory period, which dominates our training sample. While our news factorization correctly identified negative sentiment, the semantic models had limited exposure to training examples involving systemic bank runs, as such events were rare in the 1999–2023 dataset. Furthermore, the uncertainty surrounding regulatory interventions introduced path dependencies that are difficult to capture through standard news-factor frameworks.

Our second case involves Netflix after its April 2022 earnings announcement. On April 20, 2022, following the disclosure of Netflix's first quarterly subscriber loss, we forecasted a five-day decline of -6.25% based on earnings sentiment classification and factor adjustments. But the actual cumulative loss was -16.67% , producing a -10.42 percentage point error. This case illustrates the difficulty of capturing business model transitions that trigger structural valuation changes rather than marginal adjustments. The shift from high-growth to mature-company status implied a discontinuous repricing that extended beyond the linear relationships learned from typical earnings beats or misses. Our event knowledge graph successfully extracted the earnings event structure but could not fully represent the strategic implications of market saturation and intensified competition from Disney+ and HBO Max. The forward-looking guidance also introduced multi-quarter concerns that exceeded our five-day forecast window.

These two cases highlight that while MMRepAgent demonstrates strong performance under normal market conditions, extreme regime-shifting events involving nonlinear dynamics pose challenges common to models trained on central-tendency data. The primary contributing factors include: limited training samples of tail events, linear factor assumptions that may not capture threshold effects during regime transitions, and semantic understanding constraints for unprecedented event types. We note that these challenges are inherent to data-driven financial forecasting and underscore the importance of human oversight in high-stakes decision contexts. Looking forward, potential enhancements include incorporating data augmentation techniques to synthetically enrich tail event representations, exploring nonlinear modeling extensions to capture regime-dependent factor sensitivities, and integrating market stress indicators that trigger adaptive forecast adjustments. These directions aim to improve robustness during extreme conditions while preserving the interpretability and strong baseline performance that characterize MMRepAgent in typical market environments.

5. Conclusions

In this paper, we propose **MMRepAgent**, a unified framework for multi-modal financial analysis and report generation. The system integrates user requests with four key analytical tools—Stock Factorization, News Factorization, Return Forecasting, and Risk Assessment—and synthesizes results via a large language model into interpretable reports containing text, charts, and knowledge graphs. By incorporating event-aware modeling into classical financial frameworks, MMRepAgent achieves superior performance in return forecasting, risk estimation, and report quality, offering a comprehensive solution for intelligent financial decision support.

MMRepAgent, however, has limitations that merit further investigation. Its computational demands may hinder deployment on resource-constrained devices, requiring optimized architectures. Potential multicollinearity among news-derived factors, though mitigated, could be further addressed with advanced feature selection techniques. Additionally, while the model generalizes across select markets, broader dataset coverage is needed. Future work will focus on improving computational efficiency, enhancing factor orthogonality, and extending evaluations to diverse global markets to strengthen MMRepAgent's utility for retail investors.

CRedit authorship contribution statement

Xiangyu Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization; **Yawen Zeng**: Writing – review & editing, Writing – original draft, Investigation, Data curation; **Xiaofen Xing**: Writing – review & editing, Project administration; **Jin Xu**: Writing – review & editing, Project administration; **Xiangmin Xu**: Writing – review & editing, Supervision, Project administration.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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